SRMECOTECH

SRS Rules and Regulations

Standard for writing, reviewing, approving and changing Software Requirements Specifications in the GPS Processing Layer project

Document identifier	SRM-GPS-SRS-RUL-001
Version	1.0
Status	Issued for use
Date	29 October 2026
Prepared by	Sanjoy Chattopadhay, SRMECOTECH
Classification	Internal
Governing standard	ISO/IEC/IEEE 29148:2018

This standard is normative for every requirements document of the project. Where it conflicts with an institution's or client's prescribed template, that template governs and the deviation is recorded in the SRS.

Document Control

This document is placed under configuration control. Changes are made only through the change-control procedure in the companion document “SRS Rules and Regulations” and are recorded in the revision history below. A version number ending in “.0” denotes a baselined (approved) issue; any other number denotes a working draft.

Table 1: *Revision history*

Version	Date	Author	Description of change
1.0	29 Oct 2026	S. Chattopadhay	First issue: rules R1 to R12, review and approval, change control, configuration management, regulatory compliance, quality checklist and templates.

Table 2: *Review and approval*

Role	Name	Signature	Date
Product owner			
Technical lead / architect			
Quality assurance lead			

Table of Contents

Document Control 2
1 Introduction 4
1.1 Purpose 4
1.2 Applicability 4
1.3 Normative References 4
1.4 Terms Used in This Standard 4
2 Rule R1: Follow the Recognised Standard and Its Quality Characteristics 5
3 Rule R2: Use the Prescribed Document Structure 6
4 Rule R3: Write Each Requirement in Precise Language 7
4.1 Worked Example 7
5 Rule R4: Identify Every Requirement and Record Its Attributes 8
6 Rule R5: Define Scope and System Boundaries 9
7 Rule R6: Specify Functional Requirements Module by Module 10
8 Rule R7: Cover Exceptional and Edge Cases 11
9 Rule R8: Make Non-Functional Requirements Measurable 12
10 Rule R9: Document Interfaces and Data 13
11 Rule R10: Maintain Traceability from Need to Test 14
12 Rule R11: Use Diagrams Where They Add Precision 15
13 Rule R12: Apply Professional Document Control 16
14 Regulation: Review and Approval 17
14.1 Roles 17
14.2 Review Procedure 17
15 Regulation: Change Control 18
16 Regulation: Configuration Management and Storage 19
17 Regulation: Legal and Regulatory Compliance in an SRS 20
18 Final Quality Checklist 21
19 Appendices 22
19.1 Appendix A: Requirement Templates 22
19.2 Appendix B: Change Request Form 22
19.3 Appendix C: Review Record 22

1 Introduction

1.1 Purpose

This standard lays down the rules for writing a Software Requirements Specification (SRS) and the regulations for reviewing, approving, changing and storing it. An SRS written to this standard is formal, structured, testable, traceable and unambiguous: a developer can implement from it and an independent tester can verify the product against it without guessing what the author meant.

An SRS states **what** the system must do, **how well** it must do it, and the **constraints** it must satisfy. It is not a description of what the developers happen to intend to build. Architecture and technology appear in an SRS only as constraints, and only when stakeholder needs, system boundaries or recorded project decisions justify them.

1.2 Applicability

- SRM-GPS-SRS-001, Software Requirements Specification: GPS Processing Layer.
- SRM-GPS-DBS-001, Database Schema Specification (for its data requirements and change control).
- Every future SRS, addendum or module specification of the project, whether for an academic submission, a research prototype or an industry release.

1.3 Normative References

Table 3: *References*

Ref.	Document
[N1]	ISO/IEC/IEEE 29148:2018, Systems and software engineering: Life cycle processes, Requirements engineering. Primary reference. (The 2018 edition is current; a successor is in development.)
[N2]	IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications. Historical; superseded by [N1]. Its outline may be used only where it agrees with [N1].
[N3]	ISO/IEC 25010, Systems and software quality models (vocabulary for non-functional categories).
[N4]	The institution's or client's prescribed template, formatting, citation style and evaluation rubric, when one exists.

1.4 Terms Used in This Standard

Table 4: *Normative keywords*

Keyword	Meaning in this standard and in every SRS written to it
shall	Mandatory. A “shall” statement must be satisfied and must be verifiable.
shall not	Prohibited.
should	Recommended. Not following it requires a recorded justification.
may	Permitted; optional.
will	Statement of fact or of an external party's intent. Never used for a requirement on the product.

2 Rule R1: Follow the Recognised Standard and Its Quality Characteristics

R1.1 Every SRS shall follow ISO/IEC/IEEE 29148:2018 [N1] and shall state so on its title page.

R1.2 Every requirement, and the SRS as a whole, shall have the characteristics below. Reviewers shall check them using the questions in the last column.

Table 5: *Quality characteristics*

Characteristic	Definition	Review question
Correct	The requirement states a genuine stakeholder need.	Which objective or stakeholder does it come from?
Unambiguous	It has exactly one reasonable interpretation.	Would two testers write the same test from it?
Complete	All functions, constraints, interfaces and exceptional cases are covered.	Are the invalid, duplicate, late, unauthorised and dependency-failure cases stated?
Consistent	No requirement contradicts another, a figure or a table.	Do the numbers, names and states agree everywhere?
Verifiable	It can be tested, inspected, analysed or demonstrated.	What is the pass/fail criterion and the method?
Feasible	It can be implemented within the project's constraints.	Has the technical lead confirmed it?
Traceable	It has a unique ID, a source and a verification link.	Does it appear in the RTM with objective, component and test?
Modifiable	It can be changed without side effects on unrelated text.	Is it stated once, in one place, and referenced elsewhere by ID?

3 Rule R2: Use the Prescribed Document Structure

R2.1 An SRS shall contain the following parts, in this order. Headings shall be numbered consistently.

Table 6: *Mandatory SRS structure*

Part	Content	Required
Front matter	Title page, document control (revision history, approvals, distribution), table of contents.	Yes
1 Introduction	Purpose, scope (in and out), audience, definitions and acronyms, references, document conventions.	Yes
2 Overall description	Product perspective (context diagram), functions, user classes, operating environment, constraints, assumptions and dependencies.	Yes
3 Specific requirements	Functional requirements module by module, business rules, error handling and exceptional cases.	Yes
4 Non-functional requirements	Performance, capacity, reliability, availability, security, privacy, usability, maintainability, compatibility, regulatory compliance.	Yes
5 External interfaces	User interfaces, APIs, messages, data stores, hardware and communication interfaces.	Yes
6 Data requirements	Entities, validation, retention, consistency, privacy, backup and recovery (may refer to a schema document).	Yes
7 Verification and acceptance	Verification methods and measurable acceptance criteria.	Yes
8 Traceability	RTM, open issues register.	Yes
Appendices	Glossary extensions, models, configuration parameters, checklists.	As needed

R2.2 Headings that do not apply to a project may be kept with the text “Not applicable” and a reason, or removed when the institution's template allows. A heading shall never be left empty.

R2.3 Where an institution or client prescribes its own template [N4], that template shall be used and every deviation from this structure shall be listed in Section 1 of the SRS.

4 Rule R3: Write Each Requirement in Precise Language

R3.1 One requirement shall state one obligation. A statement joined by “and” or “or” that could pass for one part and fail for the other shall be split.

R3.2 A requirement shall follow the pattern: **<ID>: The <system or component> shall <action> <object> [<condition>] [<measurable criterion>].**

R3.3 The following words shall not appear in a requirement unless the same sentence or a referenced definition makes them measurable:

Table 7: *Vague words and how to make them measurable*

Vague word	Ask	Measurable form (example)
fast, quickly	How long, at what percentile, under what load?	within 1 s at the 95th percentile at 75 frames/s
real-time	What end-to-end delay is acceptable?	visible within 2 s of processing
user-friendly	Which users, which task, what success rate?	5 new operators each complete the task within 3 minutes
efficient	Which resource, what limit?	at most 64 MB of off-heap memory per RocksDB store
secure	Against what threat, by what control?	HTTP 401 for any /api call without a valid session
reliable	What failure, what tolerated loss?	no acknowledged frame lost after kill -9 of the service
sufficient, appropriate, adequate	By what criterion?	replace with the criterion itself
etc., and so on, including but not limited to	What exactly is included?	list every item

R3.4 Functional and non-functional requirements shall be kept apart. A functional requirement states a capability; a non-functional requirement states a measurable quality or constraint of a capability.

4.1 Worked Example

Table 8: *Weak versus compliant requirements*

Weak (not acceptable)	Compliant
The system should quickly show where trucks are.	FR-LIVE-001: The live-location consumer shall keep, for each IMEI, the newest position in Redis hash gps:live:vehicle:{imei}. NFR-PERF-003: At design peak load, for 95 % of frames, the vehicle's live position in Redis shall reflect the frame within 2 seconds of the frame being processed.
Bad GPS data must be handled properly.	FR-VAL-006: A frame with latitude outside −90 to 90 shall be rejected with reason “Latitude <v> is outside -90..90”.
The system must never lose data.	NFR-REL-001: No frame acknowledged by the ingest journal shall be lost because of a Kafka outage, service crash, kill -9 or power cut, provided the host's disk survives.

> **Note:** Numerical thresholds are engineering targets, not universal truths. They shall be derived from stakeholder needs and validated against the actual architecture; until then they carry the status Proposed (R4.3).

5 Rule R4: Identify Every Requirement and Record Its Attributes

R4.1 Every requirement shall have a unique, stable identifier. Identifiers shall never be re-used or renumbered; a deleted requirement keeps its row with status Deleted and the reason.

Table 9: *Identifier prefixes*

Prefix	Kind	Format
FR	Functional requirement	FR-<AREA>-nnn, AREA = 2 to 5 capital letters of the module
NFR	Non-functional requirement	NFR-<CAT>-nnn, CAT = PERF, CAP, REL, AVL, SEC, PRV, USE, MNT, CMP
IFR	Interface requirement	IFR-<IF>-nnn, IF = UI, API, KAF, RDS …
DR	Data requirement	DR-nnn
BR	Business rule	BR-nnn
REG	Regulatory or legal requirement	REG-nnn
CON	Design or implementation constraint	CON-nnn
ASM / DEP	Assumption / dependency	ASM-nnn / DEP-nnn
OBJ	Business objective	OBJ-nn
AC / TC	Acceptance criterion / test case	AC-nn / TC-<requirement ID>

R4.2 Requirements shall be presented in tables (a requirements register), not only as paragraphs. Each row shall carry these attributes:

Table 10: *Mandatory requirement attributes*

Attribute	Values
ID	Per R4.1
Statement	Per R3
Priority	Must, Should, Could (MoSCoW). “Won't (this time)” items are moved to the out-of-scope list.
Verification	T Test, I Inspection, A Analysis, D Demonstration
Status	Proposed, Approved, Baseline (implemented, awaiting verification), Verified, Future (roadmap only), Deleted. A document may use a subset and shall define it.
Source / objective	The objective, stakeholder or regulation it comes from (in the RTM).

R4.3 Approved requirements, proposed requirements and unresolved items shall never be mixed without their status. Unresolved items shall be marked TBD and listed in the open-issues register with an owner and a date.

6 Rule R5: Define Scope and System Boundaries

R5.1 The SRS shall state the problem solved, the users and stakeholders, the main capabilities, the included modules and workflows, the excluded functionality, the external systems and what each is responsible for, the assumptions and dependencies, and the operational constraints.

R5.2 Responsibilities at the boundary shall be explicit. For a fleet-tracking platform the SRS shall say, for each of the following, whether the system or an external party is responsible: receiving device data, parsing protocols, cleaning coordinates, calculating road distance, detecting stops and trips, predicting ETA, generating reports, issuing alerts, delivering notifications.

R5.3 A system context diagram shall show the boundary, every external entity and every flow across the boundary.

7 Rule R6: Specify Functional Requirements Module by Module

R6.1 Functional requirements shall be grouped by module. For each module the SRS shall state:

- the normal workflow and its trigger;
- preconditions;
- inputs (with their source and format) and outputs (with their destination);
- processing rules, including thresholds and their configuration keys;
- validation rules and the message for each failure;
- permissions (who may invoke it);
- failure behaviour (R7).

R6.2 Broad statements such as “The system shall manage vehicles” are not acceptable. A feature with several operations (register, edit, activate, assign device, deactivate, permissions) shall have one requirement per operation.

8 Rule R7: Cover Exceptional and Edge Cases

R7.1 For every module, the author shall answer each question below with a requirement or with a written “not applicable” and reason. The SRS shall contain a summary table of exceptional conditions.

Table 11: *Mandatory exceptional-case questions*

#	Question	Example of a required answer
1	What happens when input is missing or invalid?	Reject with a named reason; archive the payload
2	What happens when the database or an external API is unavailable?	Retry with back-off, never skip; or continue without the optional API
3	What happens when duplicate events arrive?	Idempotency key; second copy changes nothing
4	What happens when events arrive out of order?	Accept for history; never overwrite newer state
5	What happens when a user lacks permission?	HTTP 401/403; redirect to sign-in
6	What happens when data is stale?	Reject beyond a stated age
7	What happens when a service restarts?	Resume from committed offsets; restore state
8	What happens when several users change the same resource?	Last writer wins, audited; or optimistic locking
9	What happens at the capacity limit?	Back-pressure, lag grows, no loss within retention, alarm

R7.2 Different conditions shall not be merged into one requirement when they need different handling (for example an invalid coordinate, a duplicate, an out-of-order reading and an unavailable downstream service).

9 Rule R8: Make Non-Functional Requirements Measurable

R8.1 Every non-functional requirement shall name the condition (workload, environment), the metric, the threshold and the measurement interval. The table lists what each category shall specify.

Table 12: *What each non-functional category shall specify*

Category	Shall specify
Performance	Response-time or latency percentiles, throughput, the workload, the measurement interval.
Capacity / scalability	Concurrent users, devices or event rates supported, and expected growth.
Availability	Target percentage, measurement period, how it is measured, and exclusions (maintenance).
Reliability	Failure behaviour, tolerated data loss (RPO), recovery time (RTO).
Security	Authentication, authorisation, encryption, auditability, session controls, secret handling.
Privacy	Personal data identified, purpose, retention, access, erasure.
Maintainability	Test coverage, module rules, code-quality gates, where justified.
Usability / accessibility	Task-completion criteria with the user group and sample size; accessibility level (for example WCAG 2.1 AA).
Compatibility	Supported browsers, operating systems, protocols, integrations and their versions.
Data integrity	Consistency rules, duplicate handling, retention, recovery.

R8.2 An availability figure (for example 99.5 %) is acceptable only together with its measurement period, maintenance policy and exclusions.

R8.3 Thresholds that have not yet been validated against the architecture shall carry status Proposed and an open issue to baseline them.

10 Rule R9: Document Interfaces and Data

R9.1 For every API the SRS shall give: endpoint and method; request and response schema with required and optional fields; authentication and authorisation; validation and error responses; timeout and retry behaviour; rate limits and versioning; idempotency and duplicate-request behaviour.

R9.2 For every message or event stream: name, key, schema, ordering guarantee, delivery guarantee, retention, producers and consumers.

R9.3 For data: entities, identifiers, field types, constraints, relationships, timestamps and time zones, retention, ownership, backup and recovery. A data dictionary, ER diagram, API contract or state diagram may support the text but shall not replace a requirement that must be stated explicitly.

R9.4 When a separate schema document exists, the SRS shall reference it, and the two shall be changed together (CC.4).

11 Rule R10: Maintain Traceability from Need to Test

R10.1 The SRS shall contain a Requirements Traceability Matrix (RTM) with, for every requirement: its source objective, the design component that implements it, and the test case that verifies it.

R10.2 The RTM shall be generated from the requirement tables (not typed separately), so that it cannot drift from them.

R10.3 Before each baseline, the RTM shall be checked in both directions: every requirement has a test (no unverified requirement) and every test traces to a requirement (no unsupported feature or orphan test).

12 Rule R11: Use Diagrams Where They Add Precision

R11.1 The SRS shall include a system context diagram. It should include, where they clarify behaviour: use-case, data-flow, ER, sequence, state-machine, activity and deployment diagrams.

R11.2 Every figure shall have a number and a caption, shall be referred to in the text, and shall agree with the written requirements. Where a figure and the text disagree, the text governs and the figure shall be corrected.

R11.3 A diagram shall never replace a measurable performance, security or data requirement.

13 Rule R12: Apply Professional Document Control

R12.1 The title page shall show: title, document identifier, version, status, date, author or responsible team, classification and governing standard.

R12.2 Versions: 0.x is a working draft; n.0 is a baseline approved under Section 14; n.m (m > 0) is an approved minor revision. The status field shall say which.

R12.3 The document shall contain a revision history, a review-and-approval table, a table of contents, numbered headings, numbered and captioned tables and figures, a glossary and a references list. Acronyms shall be defined at first use.

R12.4 Terminology shall be consistent: one term per concept throughout the document (for example “reading” and “frame” are defined, and not used for anything else).

14 Regulation: Review and Approval

14.1 Roles

Table 13: *Roles and responsibilities*

Role	Responsibility
Author	Writes the SRS to this standard; answers review comments; keeps the RTM and open issues current.
Technical reviewer	Checks feasibility, consistency with the architecture and correctness of interfaces and data.
QA reviewer	Checks verifiability, acceptance criteria and the RTM.
Security / compliance reviewer	Checks security, privacy and regulatory requirements.
Product owner (approver)	Confirms the requirements meet stakeholder needs; approves the baseline.
Change control board (CCB)	Product owner, technical lead and QA lead; decides change requests after a baseline.

14.2 Review Procedure

1. Entry: the draft is complete per R2, every requirement has the attributes of R4.2, and the author has completed the checklist in Section 18.
2. Each reviewer records findings in the review record (Appendix C) with a severity: Major (wrong, ambiguous, unverifiable, missing or contradictory requirement) or Minor (wording, formatting).
3. The author resolves every finding or records why not; the reviewer closes it.
4. Exit: no open Major finding; every Minor finding resolved or accepted; open issues have owners and dates.
5. The approver signs the approval table; the version becomes n.0 and the status “Baselined”.

15 Regulation: Change Control

CC.1 After a baseline, a requirement shall be changed only through an approved change request (CR, Appendix B). Before a baseline, the author may change the draft freely, recording changes in the revision history.

CC.2 Every CR shall contain an impact analysis: affected requirements, design components, tests, schema, interfaces, effort and risk.

CC.3 Requirement status shall change only as follows: Proposed → Approved → Baseline (implemented) → Verified; Proposed or Approved → Deleted (with reason); Future → Proposed when it is brought into a release.

CC.4 Code and requirements change together. A pull request that changes behaviour covered by a requirement shall update the SRS in the same pull request; a pull request that adds a database migration shall update the Database Schema Specification in the same pull request.

CC.5 Every approved change increases the version (minor for clarifications and additions, major for changes that alter acceptance) and adds a revision-history row naming the CR.

16 Regulation: Configuration Management and Storage

- The requirements documents live in the repository folder `srs/` and are versioned with the code.
- The Word files are generated from the sources in `srs/generator/` (npm run build, then render.ps1 to fill the table of contents). The generator sources are the master copy: an edit shall be made there and the documents rebuilt, so that the RTM and the DDL appendix stay in step with the requirement tables and the migrations.
- Review comments may be made in the Word files; accepted changes shall be ported to the generator sources before the next build.
- File names: SRS_<Product>.docx, DB_Schema_<Database>.docx, SRS_Rules_and_Regulations.docx. Document identifiers never change between versions.
- Superseded versions are kept in the repository history; they are not deleted.

17 Regulation: Legal and Regulatory Compliance in an SRS

LR.1 Every SRS shall contain a regulatory section that identifies the laws, standards and third-party terms that apply to the product, states the resulting requirements with REG identifiers, and records legal review as an open issue until it is done.

LR.2 For products that process vehicle location in India, the review shall consider at least:

Table 14: *Regulations to consider*

Instrument	Typical consequence for the SRS
Digital Personal Data Protection Act, 2023	Location linked to a driver is personal data: purpose, notice, lawful basis, retention, access and erasure requirements.
IT Act 2000, s. 43A, and SPDI Rules 2011	Reasonable security practices; documented security controls.
CERT-In Directions, 28 April 2022	ICT logs kept 180 days in India; incident reporting within 6 hours.
AIS-140 (ARAI)	Device data formats and obligations where vehicle location tracking devices are mandated.
Motor-vehicle speed rules	Speed limits differ by vehicle class, road and state: limits must be configurable; events are not legal findings.
Third-party licences and terms	API terms (commercial use), map-data attribution (OpenStreetMap ODbL), open-source licences.

LR.3 The SRS shall state that regulatory requirements do not replace legal advice and shall name the person responsible for the legal review.

18 Final Quality Checklist

The author shall complete this checklist before a review (entry criterion of Section 14), giving evidence for each item. The last column records the self-assessment of SRM-GPS-SRS-001 version 0.9.

Table 15: *SRS review checklist*

#	Check	Evidence in SRM-GPS-SRS-001 v0.9	Result
Structure			
1	Purpose, scope, stakeholders, assumptions and constraints are documented.	Sections 1.1 to 1.3, 2.3, 2.5, 2.6	Pass
2	Functional, non-functional, interface and data requirements are covered.	Sections 3, 4, 5, 6	Pass
3	Figures, tables, references and terminology are consistent.	Numbered captions; glossary 1.4; references 1.5	Pass
Requirement quality			
4	Every requirement is clear, atomic, feasible and verifiable.	Requirement tables; verification column	Pass (review)
5	Every requirement has a unique and stable identifier.	Identifier scheme 1.6; RTM	Pass
6	Vague terms have been eliminated or precisely defined.	Load definitions 4; thresholds with percentiles	Pass (review)
7	Normal workflows, errors and important edge cases are specified.	Per-module requirements; Section 3.23 summary	Pass
Engineering rigour			
8	Performance and capacity requirements have measurable thresholds.	NFR-PERF-001 to 006, NFR-CAP-001 to 005	Pass; thresholds Proposed (OI-01)
9	Security, privacy, data integrity and recovery are addressed.	4.5, 4.6, 4.3, 6, REG-001 to 006	Pass
10	Requirements trace to stakeholder needs and verification methods.	Objectives table; RTM in Section 8.2	Pass
11	The requirements have been reviewed for contradictions and omissions.	Review per Section 14 pending	Open
Release readiness			
12	Acceptance criteria and verification approaches are documented.	Section 7 (AC-01 to AC-12)	Pass
13	Dependencies and unresolved decisions are explicitly recorded.	DEP-001 to 007; open issues OI-01 to OI-08	Pass
14	Document version, review status and change history are maintained.	Document control; status “Draft for review”	Pass

19 Appendices

19.1 Appendix A: Requirement Templates

Table 16: *Templates*

Kind	Template and example
Functional	FR-<AREA>-nnn: The <component> shall <action> <object> when <condition>. FR-TRP-003: When a device is silent for more than 3600 s, the open trip shall be closed at the last reading before the silence.
Non-functional	NFR-<CAT>-nnn: Under <workload>, <metric> shall be <threshold> at <percentile>, measured over <interval>. NFR-PERF-002: At design peak load, for 95 % of stored positions, stored_at − processed_at shall be at most 1 second.
Business rule	BR-nnn: <policy statement independent of any screen or API>. BR-001: A device shall be installed in at most one vehicle at any instant.
Interface	IFR-<IF>-nnn: <producer/consumer> shall <exchange> <message> via <channel> with <key/format/guarantee>. IFR-KAF-001: Any producer of raw frames shall publish the raw-frame JSON to gps.raw-device-frames with key = IMEI.
Data	DR-nnn: <entity/attribute> shall <constraint/retention/format>. DR-004: Every timestamp shall be stored as timestamptz in UTC.
Regulatory	REG-nnn: <instrument>: <obligation> shall <be met by> <product capability or operator process>.

19.2 Appendix B: Change Request Form

Table 17: *Change request*

Item	Value
CR number	CR-yyyy-nnn
Title	
Requested by / date	
Requirements affected	IDs, with current and proposed text
Reason	Stakeholder need, defect, regulation …
Impact analysis	Design components, tests, schema, interfaces, documents, effort, risk
Priority	Must / Should / Could
CCB decision	Approved / Rejected / Deferred, with date and names
Implemented in	Pull request, SRS version

19.3 Appendix C: Review Record

Table 18: *Review record*

#	Location (section / ID)	Finding	Severity	Resolution	Closed by
1					
2					
3					